Micro-Doppler Signature Simulation of Multirotor UAVs Using Ray Tracing

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2025 IEEE 25th International Conference on Communication Technology (ICCT) | 979-8-3315-8578-5/25/$31.00 ©2025 IEEE | DOI: 10.1109/ICCT67417.2025.11374154

***Abstract*—With the widespread application of multirotor unmanned aerial vehicles (UAVs) across various domains, incidents such as illegal intrusions have become increasingly frequent, highlighting the growing importance of UAVs detection. During flight, multirotor UAVs introduce micro-Doppler effects into radio frequency (RF) signals, which can be effectively used to distinguish UAVs from other targets such as birds. Most existing studies on micro-Doppler effects rely on radar technology; however, these solutions often lack adaptability in complex environments and suffer from limited generalizability. Furthermore, radar-based data collection typically requires expensive equipment and significant manpower. As a key technology in 6G communications, Integrated Sensing and Communication (ISAC) offers a promising new approach to addressing these challenges. In this study, the Sionna RT module is used to construct various simulation scenarios and obtain Channel State Information (CSI) data for analysis. The feasibility of using Sionna RT to simulate micro-Doppler effects induced by UAV rotor motion is validated. Further analysis of the micro-Doppler characteristics of multirotor UAVs reveals that, as the carrier frequency increases, the micro-Doppler spectrum becomes broader while the signal energy becomes more dispersed.**

***Index Terms*—Micro-Doppler, Ray Tracing, UAV, ISAC**

I. Introduction

The rapid development of 5G mobile communication systems and artificial intelligence technologies has sparked widespread interest in the low-altitude economy [1]. Due to their ease of deployment, low maintenance costs, high maneuverability, and ability to hover, unmanned aerial vehicles (UAVs) have found extensive applications in various civilian domains [2]. However, the proliferation of multirotor UAVs in logistics, security, and surveillance has brought about increasing risks such as illegal intrusions and privacy violations [3]. Addressing these threats requires effective UAVs detection technologies [4], making high-accuracy, low-cost UAVs detection a prominent research topic.

Nevertheless, UAVs are often difficult to detect due to their typically low flight speeds, low altitudes, small radar cross-sections (RCS), and hovering capabilities [5]. Although high radar transmit power can enhance detectability, it remains challenging to distinguish UAVs from other flying objects such as birds in urban environments, as they exhibit similar speeds and RCS [6]. However, the propellers and structural vibrations of UAVs during flight induce Micro-Doppler effects in the RF signals, which manifest as unique high-frequency modulated sidebands around the Doppler frequency [7]. These signatures enable effective differentiation between UAVs and other flying objects such as birds [6].

Currently, most Micro-Doppler-based UAVs detection studies rely on radar systems. However, there is no widely accepted open-source benchmark dataset for Micro-Doppler analysis, and radar-based data collection remains expensive and labor-intensive [8]. To address the scarcity of training data, [9] proposed the concept of channel foundation models (CFMs), which leverage self-supervised learning to exploit large-scale unlabeled data, thereby reducing the reliance on manual annotations, in [10], a mathematical model was used to derive the Micro-Doppler signatures of UAVs propellers under a bistatic OFDM waveform scenario, and Micro-Doppler detection was carried out based on the acquired channel state information (CSI), [11] have proposed several solutions, including synthetic Micro-Doppler data generation via model-based simulations, the use of generative adversarial networks (GANs), unsupervised pretraining, and transfer learning. Nonetheless, these methods are not universally applicable to all target scenarios.

Traditional theoretical models [12] and empirical models [13] typically offer computationally efficient solutions with satisfactory accuracy. However, they are generally distance-dependent and only applicable to environments similar to their development conditions [14]. Ray tracing, as a more advanced propagation modeling approach, provides more generalized solutions than empirical models [15]. Sionna RT [16], as an emerging differentiable ray tracing framework, shows potential for investigating Micro-Doppler effects in integrated sensing and communication (ISAC) systems. To date, no research has systematically employed ray tracing to model and analyze signal spectrum variations caused by rotational motions of multirotor UAVs.

The main contributions are as follows:

1) We propose an ISAC data generation method tailored for ISAC scenarios, capable of simulating UAVs targets under diverse motion patterns and parameter settings.
2) We propose an improved directional ray-tracing strategy based on Sionna RT, which replaces spherical sampling with conical sampling to enhance path density and distribution accuracy in critical regions.

3) We validate the effectiveness of Sionna RT in simulating Micro-Doppler effects induced by UAVs rotor motion. The simulation results show that the extracted Micro-Doppler features are stable and highly correlated with UAVs physical parameters, such as attitude angles, providing a basis for motion state inference.

II. Analysis of Micro-Doppler Signature of UAVs

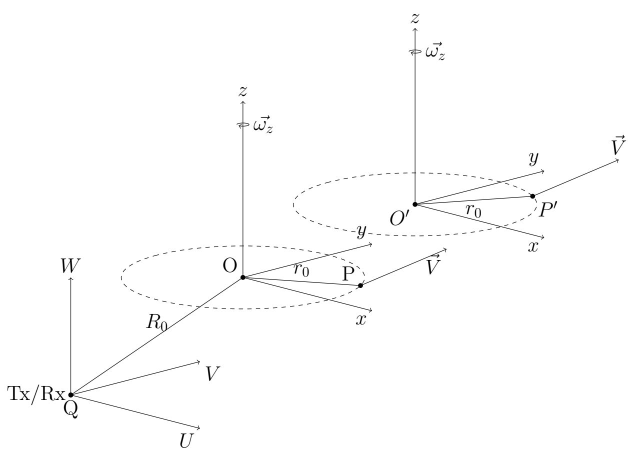


Fig. 1. Geometry of transmitter/receiver and target with translation and rotation..

As shown in Fig. 1, the transmitter and receiver remain stationary at origin Q of the transmitter coordinate system (U, V, W). The target is described in its local coordinate system (x, y, z), which undergoes translation and rotation relative to the transmitter frame. Assuming a rigid body target with translation velocity $\vec{V}$ and angular velocity $\vec{\omega} = (\omega_x, \omega_y, \omega_z)^T$ expressed in the local frame.

The rigid body motion can be described through positional changes between two time-steps: For point P on the target with initial position $\vec{r}_0$ in local coordinates at $t = 0$, its motion to P' at time t includes: (1)Global translation from O to O' with displacement $\vec{V}t$, (2)Local rotation about the origin in the body-fixed frame.

With attitude change described by rotation matrix $\mathbf{R}(t)$, the point's position in transmitter coordinates becomes:

$$\vec{r}(t) = \vec{R}_0 + \vec{V}t + \mathbf{R}(t)\vec{r}_0 \tag{1}$$

where $\vec{R}_0$ is the initial position of the local frame origin. The transmitter-to-point distance $r(t) = \|\vec{r}(t)\|$

For a transmitted sinusoidal wave with frequency f, the baseband return signal from point

$$s(t) = \frac{1}{2}\sigma(x, y, z) \cdot \exp\left\{j\frac{4\pi f}{c}r(t)\right\} \tag{2}$$

where $\sigma(x, y, z)$ is the reflectivity function and $\phi(t) = \frac{4\pi f}{c}r(t)$ represents signal phase. The Doppler shift is obtained by phase differentiation:

$$f_D = \frac{1}{2\pi}\frac{d\phi(t)}{dt} = \frac{2f}{c}\left[\vec{V} + \frac{d}{dt}(\mathbf{R}(t)\vec{r}_0)\right]^T \vec{n} \tag{3}$$

with unit direction vector $\vec{n} = \frac{\vec{r}(t)}{\|\vec{r}(t)\|}$ Using the skew-symmetric matrix representation for rotation with angular velocity $\vec{\omega}$:

$$\hat{\omega} = \begin{bmatrix} 0 & -\omega_z & \omega_y \\ \omega_z & 0 & -\omega_x \\ -\omega_y & \omega_x & 0 \end{bmatrix} \tag{4}$$

In the reference coordinate system, the angular velocity vector is described by $\vec{\omega} = (\omega_x, \omega_y, \omega_z)^T$. The object rotates around the unit vector $\vec{\omega'} = \vec{\omega}/|\vec{\omega}|$ with an angular speed of $\Omega = |\vec{\omega}|$. Assuming that the symbol duration is short and the angular velocity is relatively low, the rotational motion during each small time interval can be considered infinitesimal. Therefore, the rotation matrix can be approximated as:

$$\mathbf{R}(t) = \exp(\hat{\omega}t) \tag{5}$$

where $\hat{\omega}$ is the skew-symmetric matrix associated with $\vec{\omega}$. Therefore, the Doppler frequency shift caused by the rotation becomes:

$$\frac{d}{dt}(\mathbf{R}(t)\vec{r}_0) = \hat{\omega}\mathbf{R}(t)\vec{r}_0 = \vec{\omega} \times \vec{r}(t) \tag{6}$$

When the distance between the transmitter and receiver and the target $\left\|\vec{R}_0\right\|$, is much greater than the target's motion range $\vec{V}t + \mathbf{R}(t)\vec{r}_0$, the unit vector can be approximately considered as $\vec{n} \approx \frac{\vec{R}_0}{\|\vec{R}_0\|}$

Therefore, the final Doppler frequency shift can be approximately considered as:

$$f_D = \frac{2f}{c}\left[\vec{V} + \vec{\omega} \times \vec{r}\right]_{\text{radial}} \tag{7}$$

where $\vec{V}$ corresponds to the Doppler frequency shift caused by translation, $\vec{\omega} \times \vec{r}$ corresponds to the Micro-Doppler caused by rotation.

III. Ray Tracing Strategy

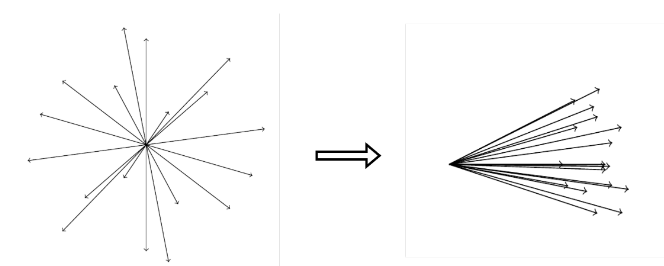


Fig. 2. Ray tracing strategy.

Under the condition of a fixed number of rays, as the distance between the target and the transmitter gradually increases, the number of rays that collide with the target gradually decreases until it approaches zero. At this point, Doppler frequency shifts can no longer be detected. Meanwhile, when the number of rays colliding with the target is small, gaps appear in the resulting time-frequency spectrogram. These gaps are not caused by insufficient Doppler frequency resolution but rather due to the scarcity of effective colliding rays

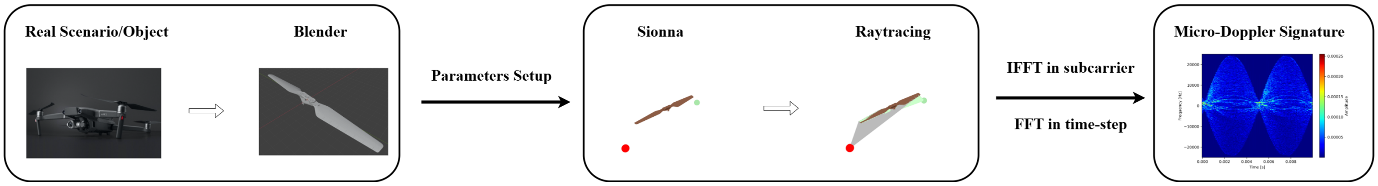


Fig. 3. Experiment diagram

To address this problem and achieve directional ray tracing toward the target, we propose a ray tracing strategy. This strategy first calculates the unit vector pointing from the transmitter to the target based on their known positions as the primary orientation. Then, a cone with this vector as its main axis is constructed, limiting the spatial range of ray emission in this direction, thereby improving the coverage efficiency of rays in the critical area, as illustrated in Fig 2. We utilize the built-in Fibonacci sampling function provided by the Sionna[1] framework to generate N uniformly distributed sampling points on the unit square $[0,1]^2$, where the sampling point indices are $i = 0, 1, \ldots, N-1$. The coordinates of the sampling points (x_i, y_i) are given by the following equation:

$$x_i = \left(\frac{i}{\phi}\right) mod\, 1, \;\; y_i = \frac{i}{N-1} \tag{8}$$

Here, mod denotes taking the fractional part. The sample points are first projected onto the surface of a cone defined in the local coordinate system (with the local z-axis as the main axis):

$$\mathbf{v}_{local} = \begin{bmatrix} \sin\theta\cos\phi \\ \sin\theta\sin\phi \\ \cos\theta \end{bmatrix} \quad \text{where} \begin{cases} \cos\theta = 1 - S_x(1-\cos\theta) \\ \sin\theta = \sqrt{1-\cos^2\theta} \\ \phi = 2\pi S_y \end{cases} \tag{9}$$

The polar angle θ is determined by S_x, and the azimuthal angle ϕ is determined by S_y, uniformly distributed over $[0, 2\pi]$:

When mapping the cone direction vector from the local coordinate system to the global coordinate system, we use the Frame utility function from the Mitsuba[2] rendering framework to construct the rotation transformation $\mathbf{R}$. The direction vector in the world coordinate system $\mathbf{v}_{world}$ is obtained through the following transformation:

$$\mathbf{v}_{world} = \mathbf{R}\mathbf{v}_{local} \tag{10}$$

This ensures that the direction vectors are uniformly distributed within a cone centered around the unit direction, thereby achieving directional sampling of the target area. This strategy guarantees that most rays interact with the target region, effectively improving the observability of the target response in the simulation and providing high-quality data support for subsequent Micro-Doppler feature modeling.

Table I. Parameters Setup

RT Parameters	Value
Waveform	OFDM-based
Number of OFDM symbols	2048
OFDM symbol duration	$50\mu s$
Rotating propellers	1
Blades per propeller	2
Propeller radius	$10.55cm$
Propeller material	Wood

IV. Experiments

In this section, we validate the accuracy of the Sionna RT framework for simulating rotational micro-Doppler effects. The simulated results are compared with the analytical formulas derived in Section II using identical target models and micromotion parameters. The overall procedure, illustrated in Fig. 3, consists of three steps: physical modeling, parameter setup, and data processing. In the physical modeling, both built-in objects in Sionna and rotor models created in Blender are employed. Details of the parameter configuration are summarized in Table I.

The transmitted signal, target, and received signal are considered. The transmitted signal can be a pulse waveform, OFDM waveform, or other defined waveforms. The target consists of either Sionna's built-in reflector model or a physical model of a propeller modeled using Blender. The model's coordinates are located at its geometric center, and electromagnetic parameters such as conductivity are based on Sionna's built-in settings. The initial positions, velocities, and trajectories of the transmitter, receiver, and target are predetermined as inputs for the simulation.

In summary, the simulation process includes:

(1) Select transmission parameters (carrier frequency, bandwidth, number of subcarriers, number of symbols);

(2) Choose the physical model (Blender modeling or Sionna built-in models);

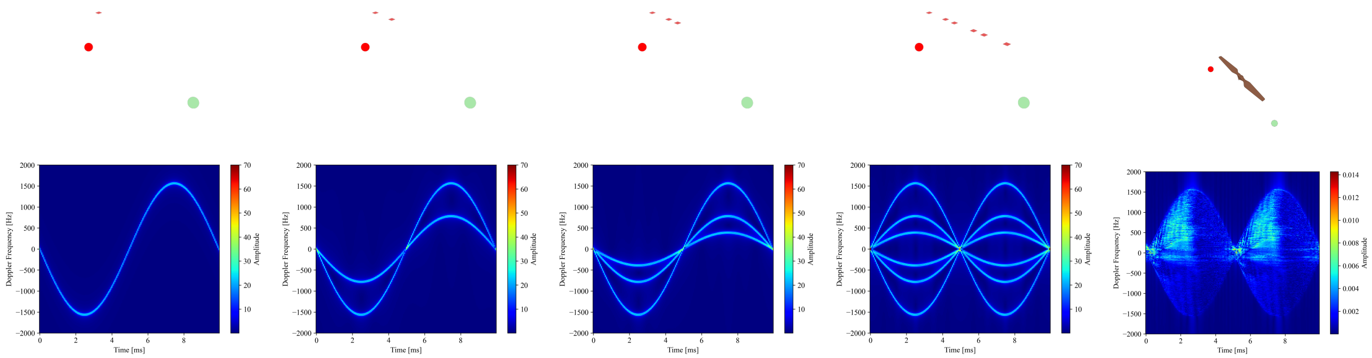
(3) Select the target motion parameters (initial position, trajectory, velocity, and acceleration of the target);

(4) Emit rays toward the target and calculate the channel frequency response;

(5) Update the target's velocity and position, then repeat step 4;

(6) Convert the computed channel data from frequency domain to delay domain using Inverse Fast Fourier Transform

[1]Sionna:https://nvlabs.github.io/sionna/
[2]Mitsuba:https://www.mitsuba-renderer.org/



(a) (b) (c) (d) (e)

Fig. 4. Geometry and Micro-Doppler time-frequency spectrum of scatter points(a):number of scatter points = 1.(b):number of scatter points = 2.(c):number of scatter points = 3.(d):number of scatter points = 6.(e)scatter point = $\rightarrow \infty$(Blade model)

(IFFT), and from time domain to Doppler domain using Fast Fourier Transform (FFT).

A. Simulation Validation of Micro-Doppler Effects Using Sionna RT

Rotation is a fundamental micromotion of the target. Assume the system operates at 5,GHz, with initial Euler angles ($\psi = 0°, \theta = 0°, \phi = 0°$), angular velocity vector $\vec{\omega} = [\omega_x, \omega_y, \omega_z]^{\mathrm{T}}$, and velocity vector $\vec{V} = [V_x, V_y, V_z]^{\mathrm{T}}$. We adopt an approach inspired by the micromechanical method, using a strategy from local to global. First, we simulate the case of a single scattering center. Then, the number of scattering centers is gradually increased—from two points, three points, six points—to finally importing a realistic blade model for simulation.

In the case of a single scatterer, the simulated Micro-Doppler modulation is shown in Fig. 4a. The modulation from a single scatterer center appears as a sinusoidal curve symmetric about zero frequency, which agrees with [7]. According to Eq. (7), the calculated maximum Micro-Doppler frequency shift is 1562 Hz. The rotation period derived from the angular velocity, $T = \frac{2\pi}{\omega} = 0.04$ s, matches perfectly with the simulation results.

For scatter points equal to 2, 3, and 6, when the angular velocity $\vec{\omega}$ remains constant, if the rotation radius r decreases, the linear velocity decreases accordingly, which leads to a reduction in the amplitude of the sinusoidal Micro-Doppler modulation. When the number of scattering centers increases and the rotation radius of the additional scattering centers decreases, the time-frequency diagram gradually fills the spectrum within the envelope of the maximum frequency shift sinusoid. The simulation results are shown in Fig. 4b, 4c, and 4d, which are consistent with the theoretical results.

When the rotating target is a physical model of a propeller, considering that the minimum spatial resolution in Sionna is 0.01 m, the number of scattering centers can be regarded as approaching infinity. According to the previous analysis, the Micro-Doppler modulation should then appear as two symmetric filled sinusoidal envelopes around zero frequency. The simulation results are shown in Fig. 4e, which conform to the theoretical expectation.

It is worth noting that when the range resolution is smaller than the physical size of the target, which means in the high range resolution (HRR) scenario, the target is no longer regarded as a point scatterer, but rather forms an extended response along the range dimension. This differs from the classical model in [17], which assumes the target response is contained within a single range cell, producing only one clear sinusoidal curve. In the HRR case, the target can be resolved into multiple independent scatterers, and the micro-Doppler spectrum appears as a filled sinusoidal envelope.

B. Impact of Carrier Frequency and Azimuth Angle on Micro-Doppler Features

In radar systems, the Micro-Doppler signal generated by rotor blades is influenced by multiple parameters, including blade length L, rotational angular velocity ω, carrier frequency f and azimuth angle. We focus on analyzing the impact of azimuth angle and carrier frequency f on the Micro-Doppler signatures generated by the rotor.

To analyze the impact of the azimuth angle on the Micro-Doppler effect, the results shown in Fig. 5 assumes a rotor with two blades rotating at a constant angular velocity, which indicates that when only the yaw angle changes, the maximum micro-Doppler frequency shift stays constant, with only the initial phase varying. However, when the pitch or roll angle changes, the maximum micro-Doppler frequency shift also changes.

Furthermore, Fig. 6 investigates the impact of the carrier frequency. Under the conditions of a blade length of 1.055 meters, constant rotational speed, and fixed azimuth angle, the range of Micro-Doppler frequency shifts increases linearly

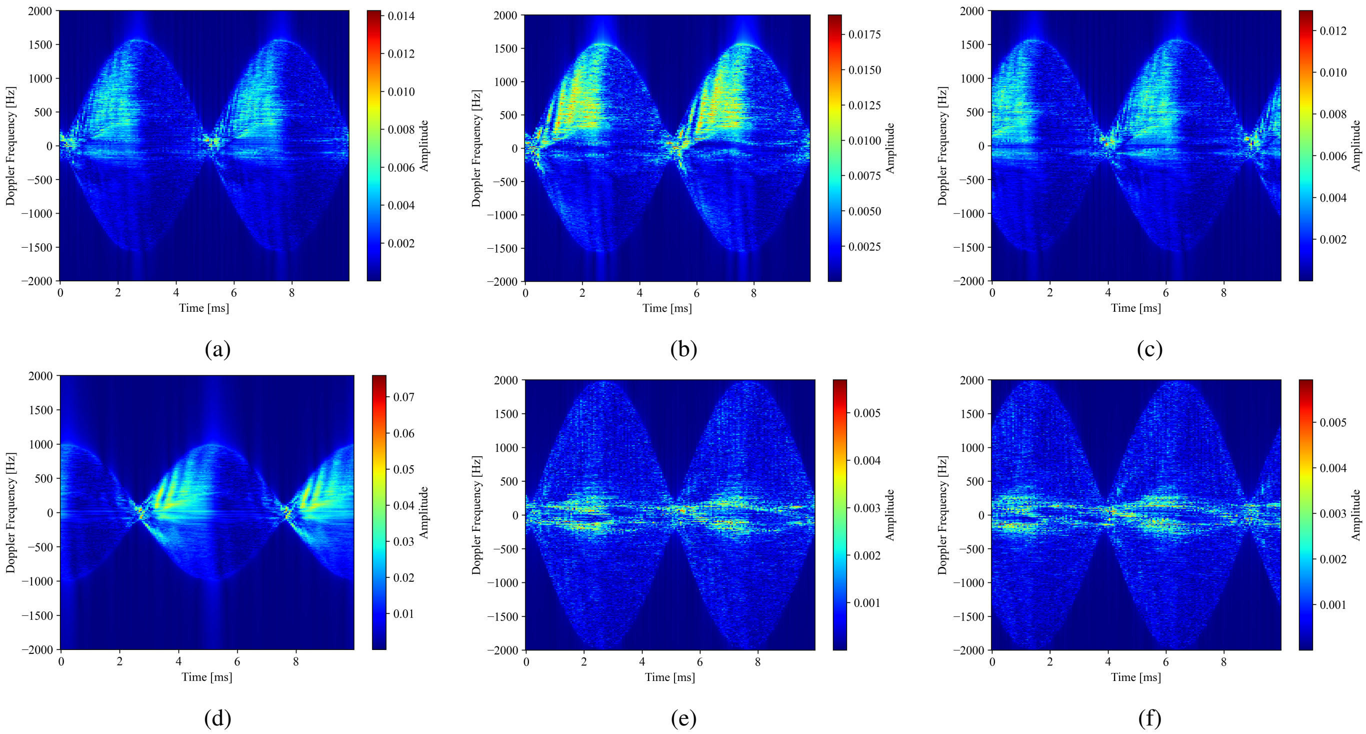


Fig. 5. Micro-Doppler signature of the different positions(Euler Angles).(a)Transmitter/Reciver at (10,0,-10).(b)Transmitter/Reciver at (5,0,-5).(c)Transmitter/Reciver at (7.071,7.071,-10).(d)Transmitter/Reciver at (5,0,-10).(e)Transmitter/Reciver at (10,0,-5).(f)Transmitter/Reciver at (7.071,7.071,-5).

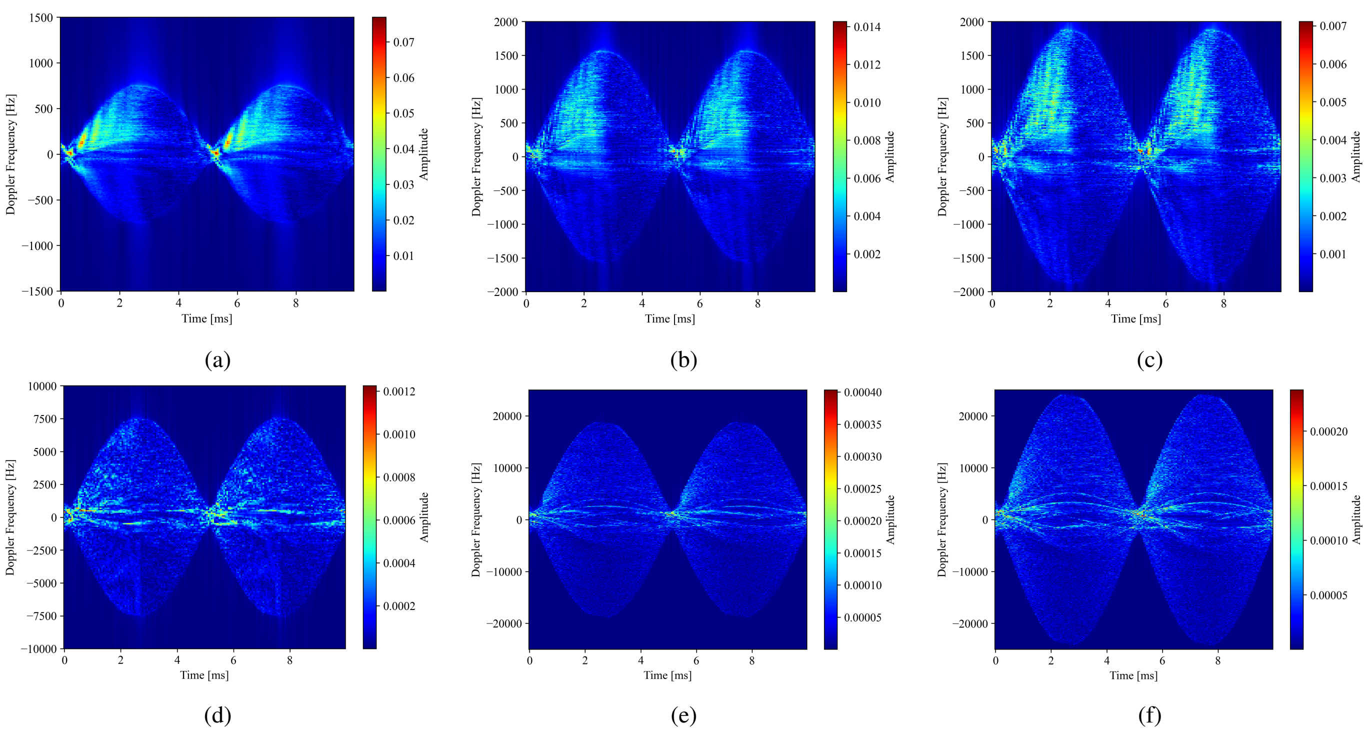


Fig. 6. Micro-Doppler signature of the different carrier frequency.(a)2.4GHz.(b)5GHz.(c)6GHz.(d)24GHz.(e)60GHz.(f)77GHz.

with the carrier frequency, as indicated by Eq. (7). However, simulation experiments also show that the echo energy of high-frequency signals significantly decreases. This reduction may be caused by atmospheric absorption. According to the free-space path loss formula, higher frequencies suffer greater path loss, resulting in lower received energy. Therefore, in practical system design, a trade-off must be made when selecting the carrier frequency to balance the amplitude of Micro-Doppler frequency shifts and the echo energy.

V. Conclusion and Future Work

This paper proposes a directional modeling strategy based on ray tracing, designed to achieve targeted ray propagation toward specific objects. By incorporating this strategy into simulation experiments, we demonstrate the feasibility and effectiveness of Sionna RT in capturing the Micro-Doppler effects caused by a single UAV's rotor motions. Simulation results show that, on one hand, as the number of scattering points approaches infinity, the periodic rotation of the rotor induces Micro-Doppler features that manifest as filled sinusoidal envelopes. On the other hand, the periodic motion of the rotor introduces stable Micro-Doppler signatures into the received signals. These signatures can not only be effectively extracted, but also exhibit strong correlations with various physical parameters of the UAV platform, such as attitude angles. Furthermore, as the carrier frequency increases, the Micro-Doppler signatures undergo greater spectral broadening, while the associated signal energy decreases.

In recent years, to address severe Doppler shifts in high-mobility communications, researchers have proposed novel techniques such as OTFS modulation, which alleviates the Doppler sensitivity of OFDM by mapping information into the delay–Doppler domain [18]. Meanwhile, deep learning-based joint channel estimation and data detection has demonstrated promising robustness, providing new perspectives for signal processing in dynamic channels. Studies on Integrated Localization and Communication (ILAC) reveal the fundamental trade-off between communication and localization performance [19] [20]. Their combination can bridge Sionna RT-based signal simulation with system-level optimization, promoting the deployment of ISAC in 6G low-altitude economy scenarios. In addition, the issue of channel aging in high-mobility UAV scenarios can be mitigated by lightweight Transformer-based channel prediction methods [21], while CSI extrapolation techniques, such as CANet, can reduce acquisition overhead and enhance detection robustness [22].

In summary, future research may advance UAV micro-Doppler detection from physical-layer modeling to system-level optimization through channel prediction , and high-resolution CSI acquisition , laying a solid foundation for practical 6G ISAC applications.

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